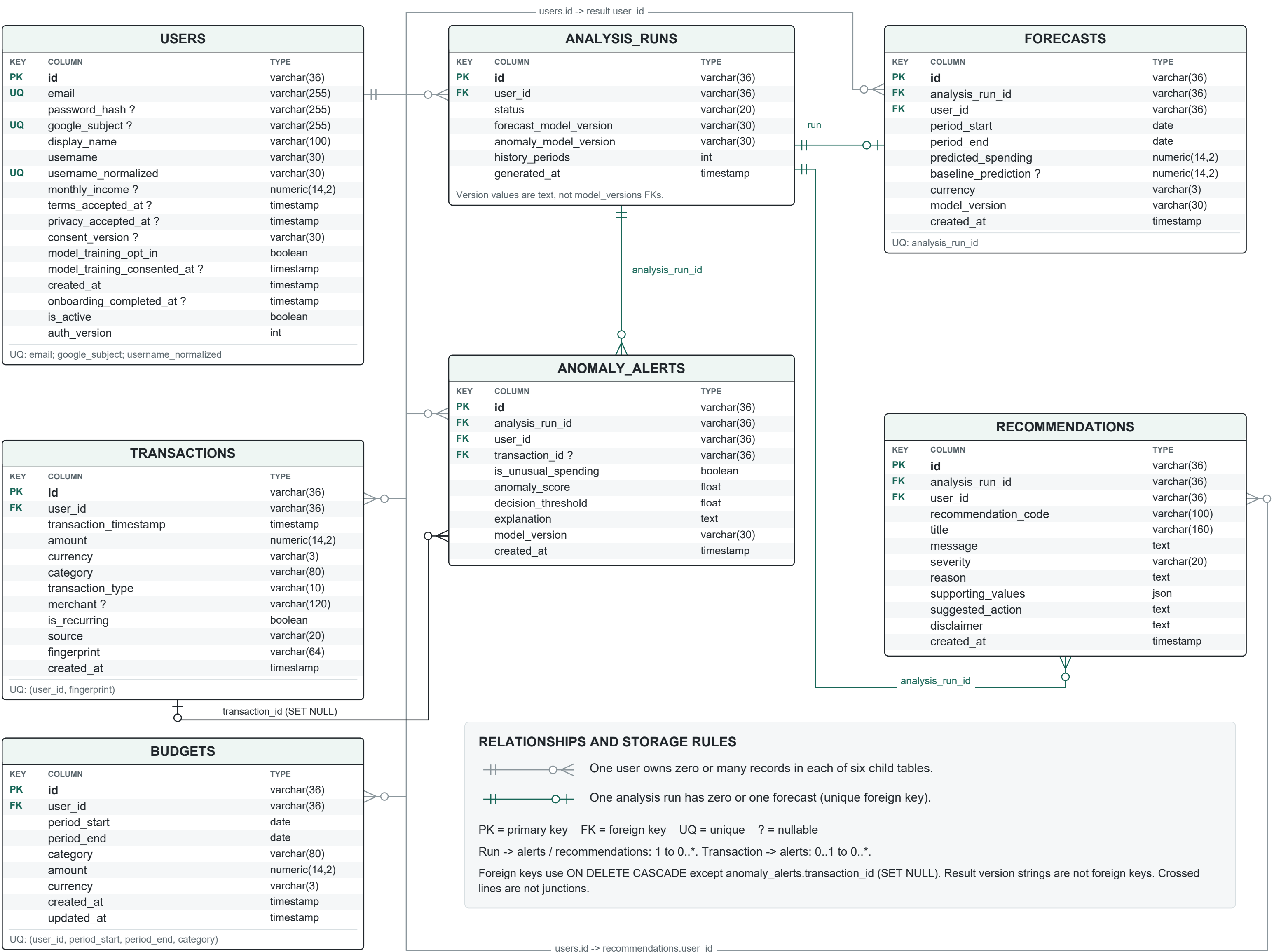


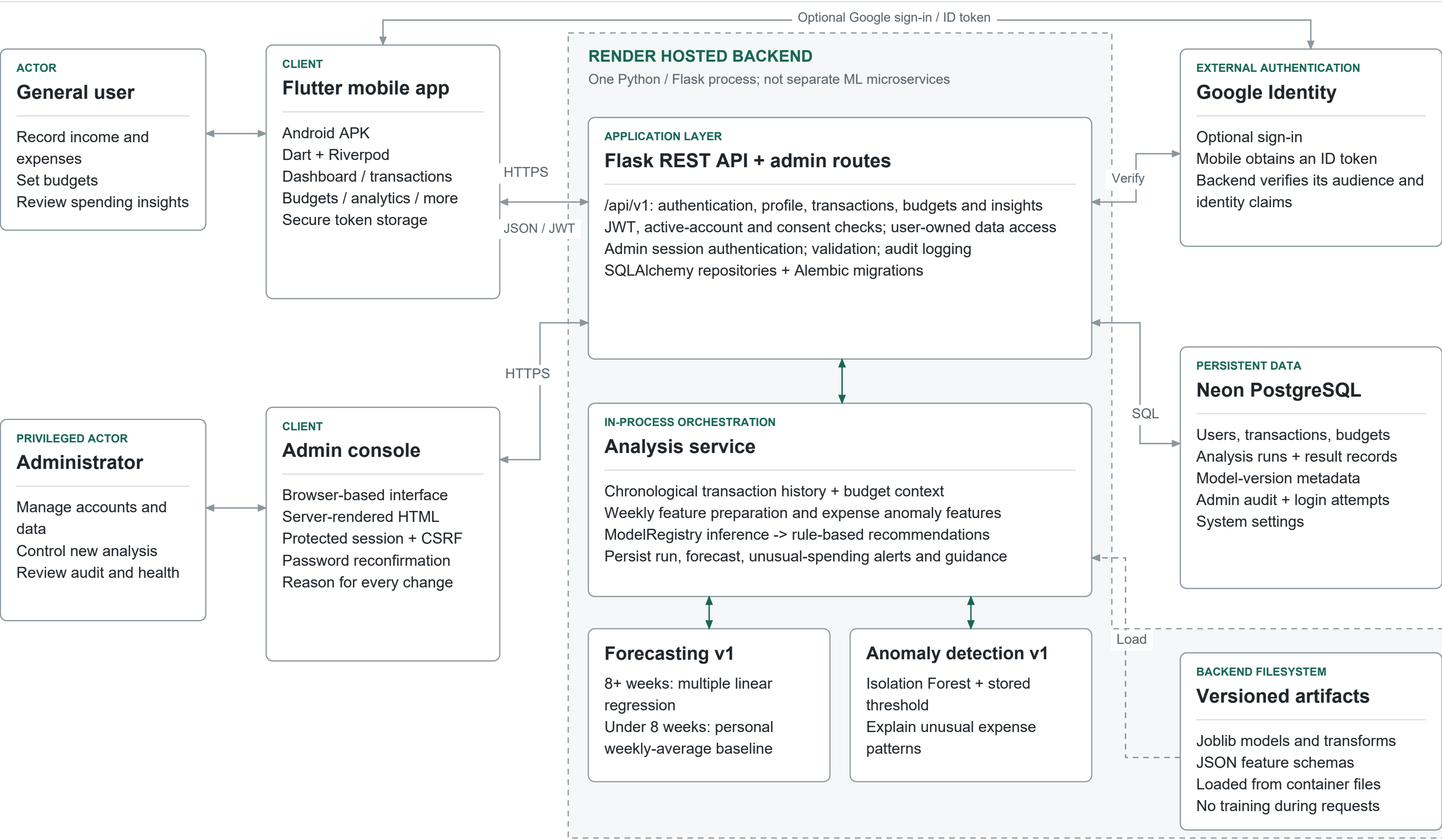
Database schema

Physical application schema | PostgreSQL | 11 tables, 97 columns and 10 foreign-key relationships



System architecture

Deployed mobile-first setup | Android client, browser admin, Flask backend and PostgreSQL storage

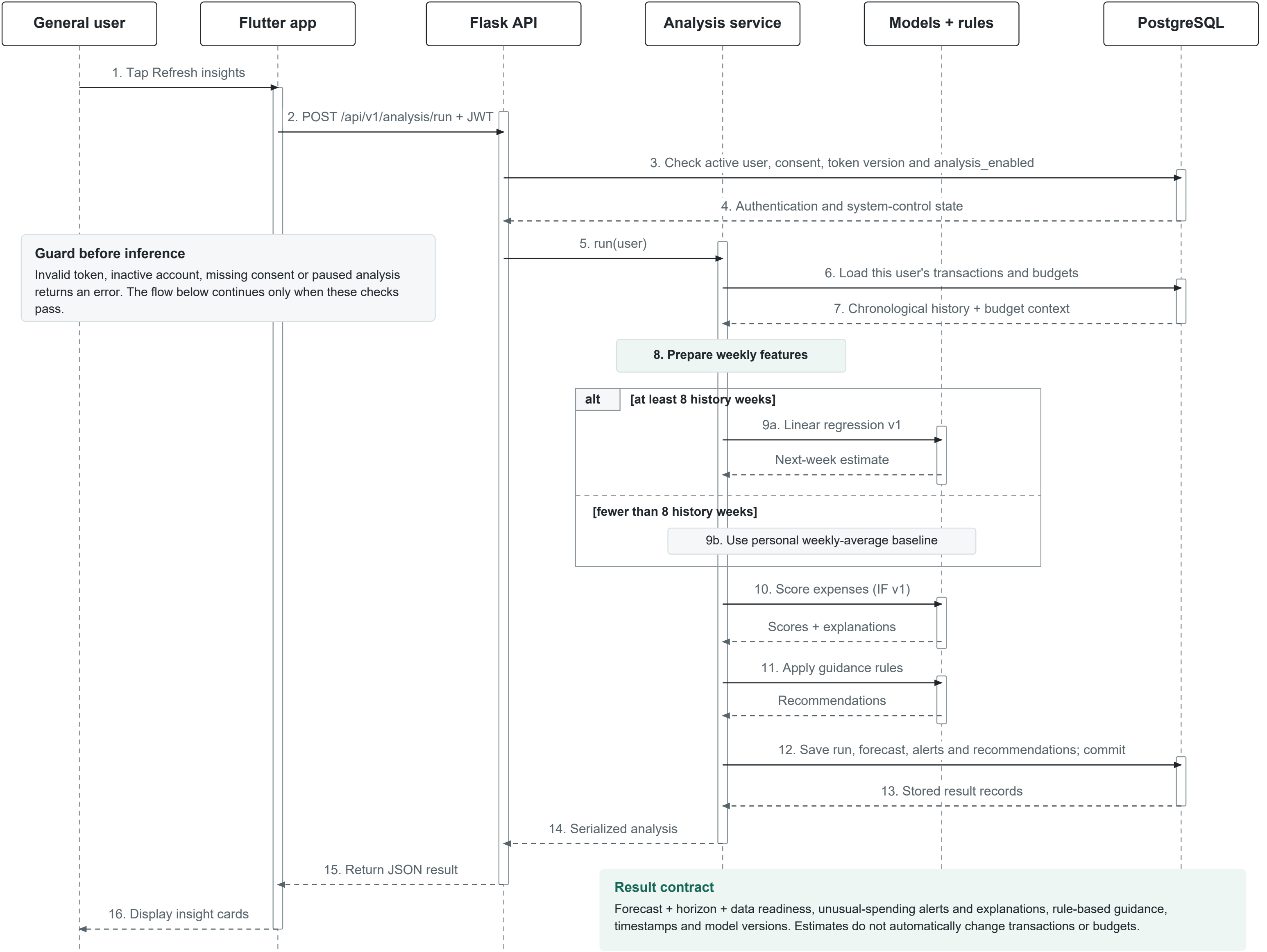


LIVE / RESEARCH BOUNDARY

Live: MODEL_RUNTIME=lightweight, forecast v1 and anomaly v1. LSTM is available only in the optional full runtime, not this deployment. Frozen V2 research candidates remain outside the live request path; integration requires a reviewed model package and adapter.

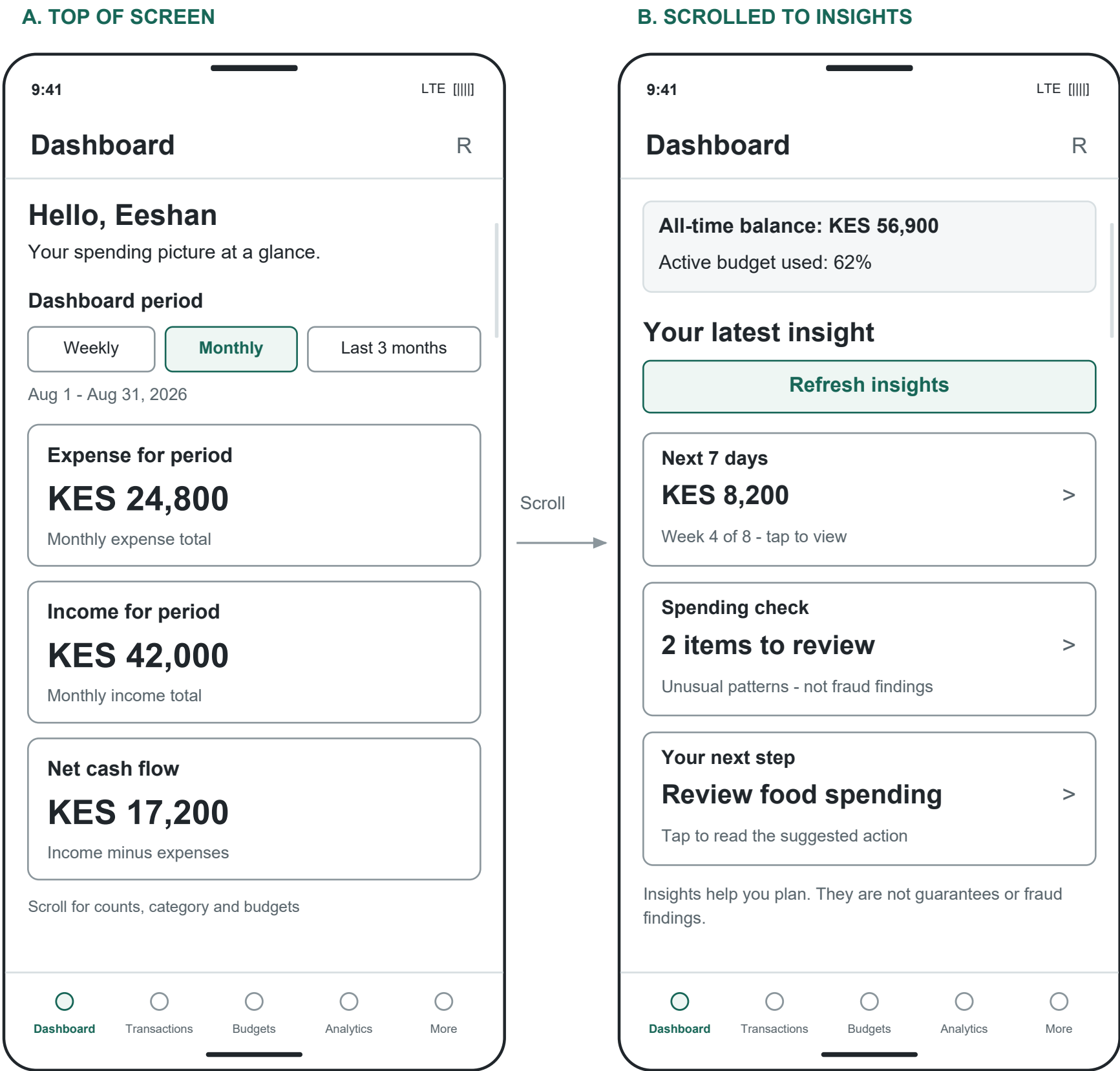
Sequence diagram - generate spending insights

Successful user flow with forecast-readiness branches | Internal calls are within the same Flask backend



Mobile wireframe - dashboard

Main financial overview | Two scroll positions of the same screen, not two separate wireframes



Mobile wireframe - add transaction

Primary data-entry flow | Expense or income, amount, category, source and transaction date

9:41LTE [||||]

< Add transaction

ExpenseIncome

Amount (KES)

850.00

Category

Food

Merchant or source (optional)

Example Store

Recurring transaction

Mark a repeated payment

Transaction date

31 Aug 2026

Save transaction

- 1

Reached from Transactions

The add action opens this page. Back returns to history; the five-tab bottom bar is not shown on this pushed page.
- 2

Clear input controls

Amount is numeric; category is a text field. Income and expense are mutually exclusive. Merchant or source is optional.
- 3

Recurring is a label

The switch tags a recurring payment. It does not create a payment schedule or initiate any bank transfer.
- 4

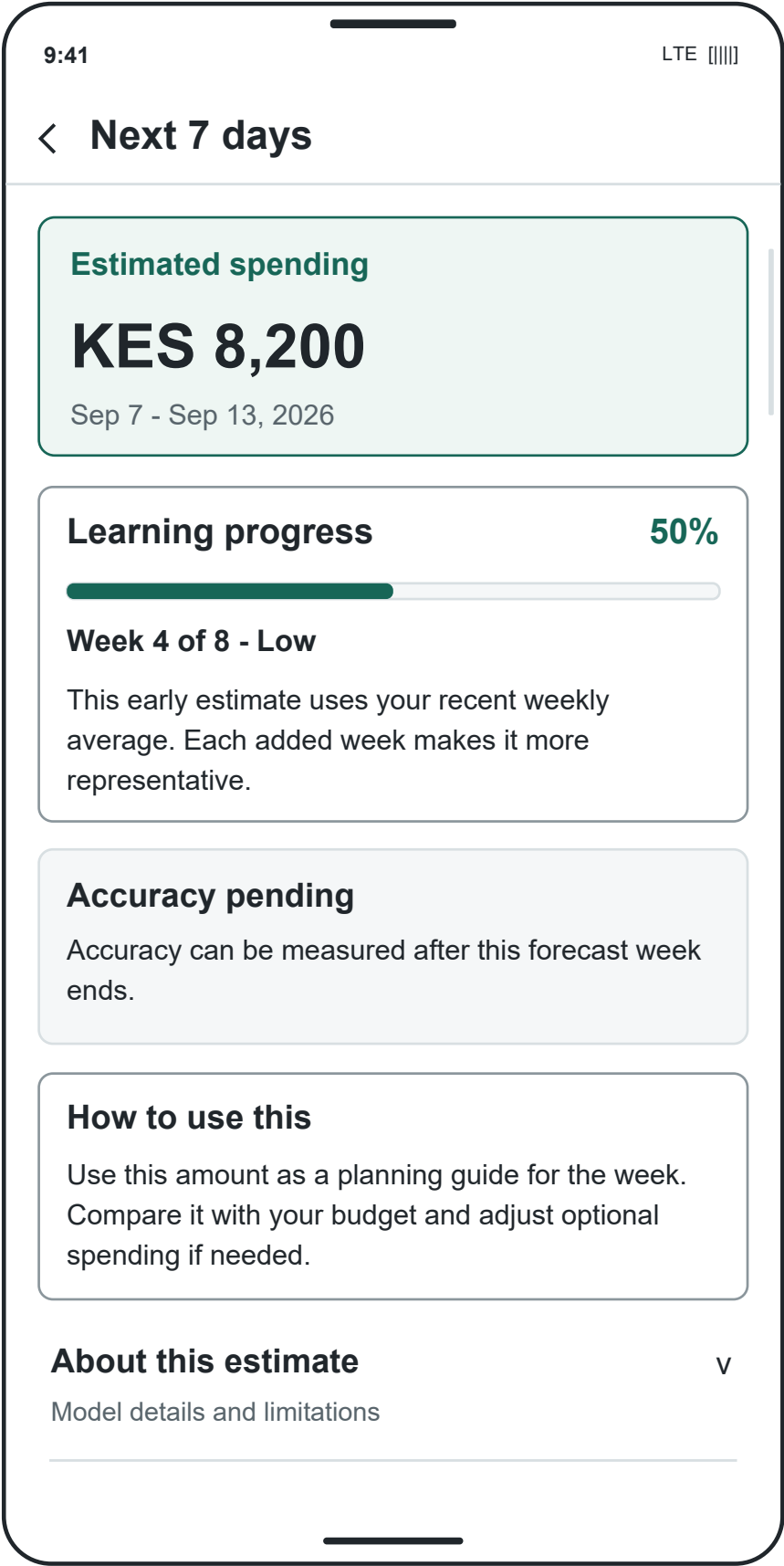
Validate, save, return

Inline errors explain invalid values. Save is disabled while submitting; success returns to transaction history.

Illustrative values. This page records financial data; it does not move money.

Mobile wireframe - forecast results

Next 7 days | Early-history example with honest data-readiness and accuracy messaging



- 1

A concrete forecast horizon
The estimate is linked to a specific seven-day period. The example here is fictional, not a claim about model performance.
- 2

Readiness is not accuracy
50% means four of eight history weeks are available. It does not mean the forecast is 50% accurate or has 50% statistical confidence.
- 3

Explain the live method
Under eight history weeks, the current backend uses the personal baseline. With enough history, the lightweight runtime uses linear regression v1.
- 4

Limitations stay visible
The details section explains synthetic-data validation and experimental status. The forecast is a planning aid, not a guaranteed outcome.

No accuracy percentage is invented for a forecast whose target week has not ended.

Admin wireframe - system management

Browser-based administration for the mobile system | Account controls, data management, system switch and audit

SPENDLY

Administrator console

Overview / model health

Account management

System controls

Activity / audit

System management

Manage access, inspect records and control new analysis requests.

Sign out

Accounts

Search name, username or email

All statuses v

Search

User	Email	Sign-in	Status	Action
Sample User / sample_user	sample@example.com	Password	Active	Manage

1 selected account

Previous1Next

Sample User

sample_user | sample@example.com | Active | Training opt-in: No

Transactions

124

Budgets

3

Income

KES 42,000

Expenses

KES 24,800

Edit profile

Disable account

Revoke sessions

Export account data

Delete account*

*Disable the account first. Sensitive changes require a reason and admin-password confirmation.

Records

Transactions v

Add record

31 Aug 2026 | Expense | Food | Example Store | KES 850

Edit

Delete

System controls

Fresh analysis: Enabled

Pause fresh insights

Lightweight runtime | Forecast v1 | Anomaly v1

The switch blocks new analysis, not in-flight work.

Activity / audit

Time	Actor	Action	Target	Reason
14:20	admin	profile update	sample user	correction

Audit records survive deletion of the target account.

POWERFUL, ACCOUNTABLE ADMINISTRATION

Record selector: transactions, budgets, forecasts, alerts and recommendations. Create applies to transactions and budgets; other supported actions follow the selected record type. Admin identity and consent are protected; existing analysis results are not silently recomputed.

Sources: backend/app/templates/admin/manage.html; routes/admin_management.py | All displayed account data is fictional.

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